PROCESS IMPROVEMENT · DMAIC CASE STUDY

Enterprise Learning Quality Transformation

Analytics • Automation • Process Improvement • Operational Excellence

ROLE Project Lead | Data Analytics | Learning Technology

Designed and implemented an automated quality management system that transformed a reactive learning-content correction process into a proactive, data-driven quality framework - serving a global audience of over 100,000 people.

AT A GLANCE

25,413+	108	1,615	100K+
Learning Objects Analyzed	Learning Object Owners Enabled	Hidden Defects Surfaced	Audience Impact (employees & partners)

2+ YEARS

SUSTAINED PERFORMANCE

Backlog reduced to near-zero

and held - while thousands more objects were added.

THE BUSINESS CHALLENGE

Training content was maintained by distributed teams with no automated validation in place, so defects accumulated silently over years - inconsistent standards, recurring errors, and costly rework that no one could see or measure. Issues spanned spelling, incorrect product terminology, missing information, punctuation, and pricing inconsistencies, all degrading learner experience, search accuracy, and operational efficiency.

THE DMAIC APPROACH

A structured Define-Measure-Analyze-Improve-Control framework drove the transformation.

D

DEFINE

Exposed defects that had accumulated undetected over years across distributed teams - hidden issues hurting learner experience, search accuracy, and efficiency.

M

MEASURE

Built automated measurement on enterprise LMS data via Power Query, Excel, Power Automate, SharePoint & Power BI.

A

ANALYZE

Found root causes: copy/paste propagation, inconsistent product naming, and no automated validation.

I

IMPROVE

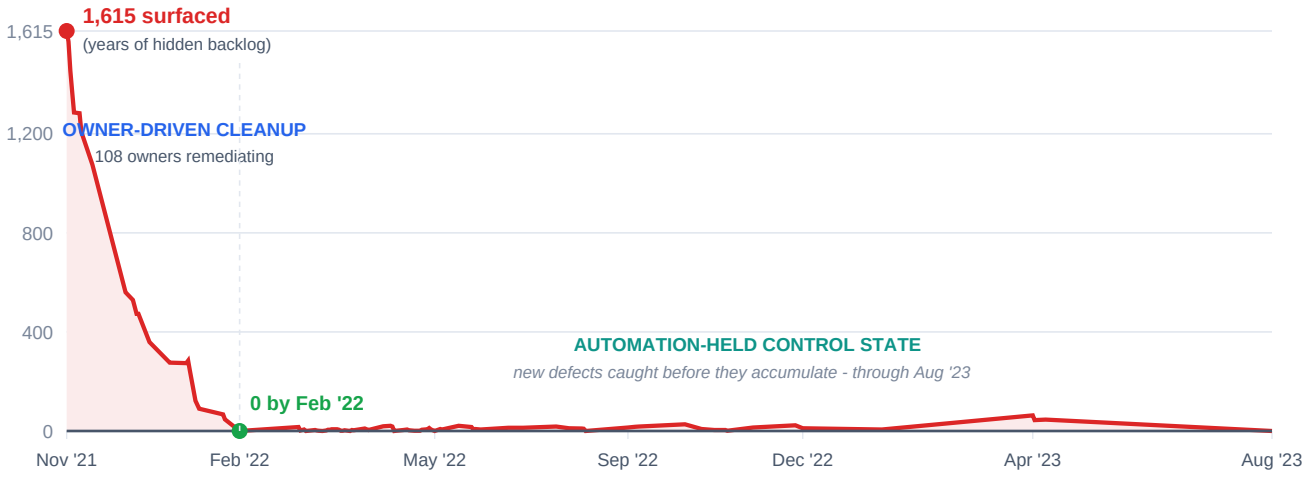
Enabled 108 owners to remediate the backlog via owner-specific reports and coaching, backed by automated detection, fuzzy product matching, and suggested corrections.

C

CONTROL

Established dashboards, daily monitoring, ownership accountability, standard work & continuous improvement.

HERO METRIC - DEFECT REDUCTION OVER TIME



Owners cleared 1,615 surfaced defects to zero in about 3 months with targeted reports and coaching. Automation then held the line for 2+ years.

BEFORE → AFTER

A reactive, defect-prone workflow was re-engineered into a proactive, self-correcting system.

BEFORE - REACTIVE

- Create Learning Object
- Copy Existing Content
- Publish
- Customer Finds Issue
- Manual Investigation
- Correction

- X Reactive - issues found by customers
- X Hidden, propagating defects
- X Repetitive manual rework

AFTER - PROACTIVE

- Daily LMS Extract
- Automated Validation
- Owner Notification
- Correction / Exception
- Dashboard Monitoring

- + Proactive - caught before publish
- + Preventative, automated checks
- + Sustainable & self-monitoring

LEAN PRINCIPLES APPLIED

Kaizen
Employee-driven improvement identification

Visual Management
Power BI dashboards & owner reporting

Standard Work
Documentation, training & quality expectations

Poka-Yoke
Automated detection & suggested corrections

Respect for People
Owners enabled through coaching & support

Waste Reduction
Less manual searching, rework & investigation

TIMELINE & RESULTS

From baseline to a sustained, governed quality state.

Nov '21	Q1 '22	Q1 '22	Apr '22	'21-'23
DEFINE + MEASURE	**ANALYZE**	**IMPROVE**	**RESULT**	**CONTROL**
Surfaced 1,615 hidden defects	Root causes identified	108 owners enabled	Owners cleared to zero	Automation held 2+ yrs

SOLUTION DESIGN - THE AUTOMATED LOOP

Daily LMS Extract > **Automated Validation** > **Owner Reports** > **SharePoint Distribution** > **Power Automate Notifications** > **Correction / Exception Loop**

RESULTS & BUSINESS IMPACT

Surfaced 1,615 defects that had accumulated undetected over years, then enabled 108 owners to clear the backlog to zero within roughly three months through targeted reports and coaching. Automation held the line thereafter - new defects stayed controlled even as thousands of additional Learning Objects were introduced across a multi-year period. The initiative created a repeatable governance model combining analytics, automation, ownership, and continuous improvement.

PROCESS CAPABILITY

3.8 SIGMA
Baseline
10,592 DPMO • Nov 2021

->

5.1 SIGMA
Sustained
153 DPMO · Feb 2022 - Aug 2023

Basis: 6 defect opportunities per course (six independent checks). Sustained sigma = 650 defects across 4.26M opportunities over 71 readings in the controlled period.

1,615 → 0	2+ yrs	100K+	108
Hidden defects cleared	Performance sustained	Audience protected	Owners enabled

A reactive correction process became a proactive, data-driven quality framework - measurable, automated, owned, and built to last.

HOW IT WORKED - THE OWNER REPORT

Each owner received a personalized report listing only their courses, the specific errors found, and suggested fixes.

ERRORS BY OWNER - DAILY VALIDATION DASHBOARD

6 automated checks per course	**Daily** refresh from LMS extract	**Per-owner** individualized reports

INDIVIDUALIZED ERROR REPORT (recreated sample)

Training Title	Error Description	Suggested Correction	Checked?	Add Words
Cloud Networking Basics | CN-1010	Title Errors: Check spelling - Kubernets	Title: Kubernets -> Kubernetes	Yes	Kubernetes
Intro to Container Orchestration | CO-2204	Desc Errors: Repeated word 'the the'; Check spelling - Dockr	Remove duplicate 'the'; Dockr -> Docker	Yes	Docker
Network Routing Fundamentals | NR-1502	Desc Errors: Missing description; UK spelling 'optimise'	Add description; optimise -> optimize	Yes	-
Firewall Configuration Part 2 | FC-0307	Title Errors: Product name - 'fortigate' inconsistent	Standardize product name -> FortiGate	No	FortiGate

THE FEEDBACK LOOP

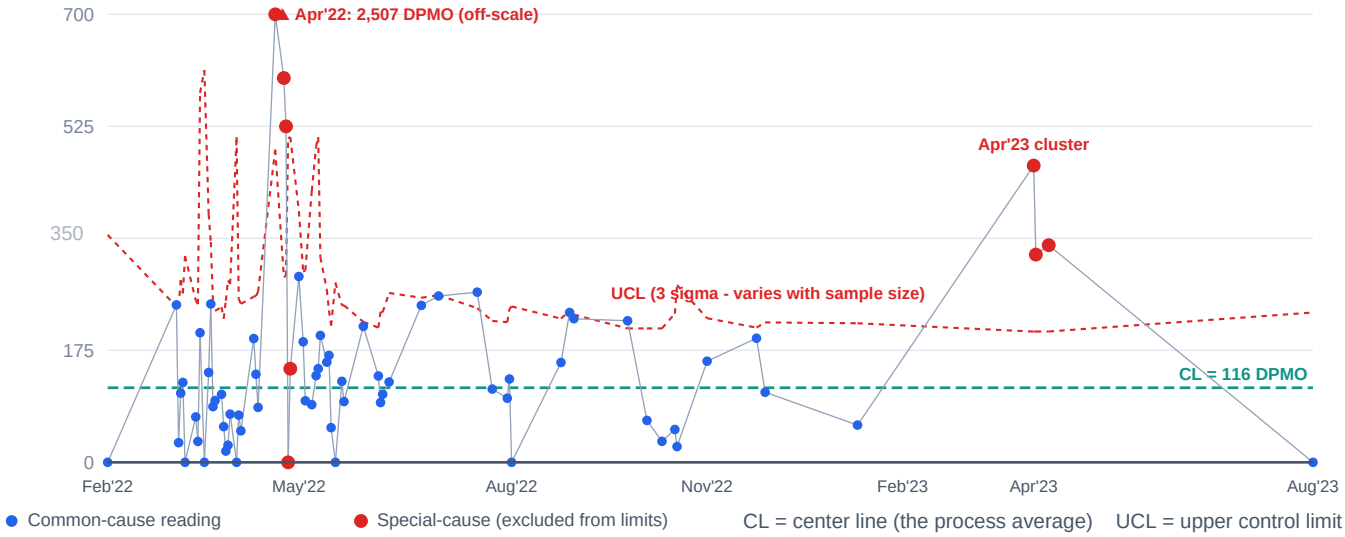
Automated daily scan > **Errors flagged + suggested fix** > **Owner corrects & marks 'Checked'** > **New terms added to dictionary** > **Cleared item drops off report**

Illustrative recreation. Table structure reflects the actual validation report; rows use fictional, generic content. No proprietary or identifying data is shown - the live system ran against an internal enterprise LMS catalog of 25,000+ learning objects.

STATISTICAL PROCESS CONTROL

Defect rate tracked on a u-chart (defects per opportunity), with control limits adjusted for each reading's sample size.

DPMO (defects per million opportunities)



The process holds in statistical control at about 117 DPMO (5.2 sigma). The two flagged clusters (Apr 2022, Apr 2023) are documented special-cause events - large annual catalog loads introduced new defects, which the system detected and owners cleared within weeks. Limits were computed from common-cause readings only, per standard practice.

MEASUREMENT SYSTEM NOTES

Opportunity basis: 6 independent checks per learning object (product names, spelling, duplicate words, punctuation, missing fields, UK-vs-US spelling). The detector had known false-positive modes - chiefly formatting artifacts and non-English titles - which were identified and addressed by deprecating formatting errors and adding a 'checked' exception path for valid non-English content. Sigma levels apply the conventional 1.5σ long-term shift.

COST OF POOR QUALITY (LABOR)

135 hrs

of latent remediation work in the surfaced backlog (1,615 defects x 5 min), invisible until detection existed.

≥ 2x faster

per fix: bundling the flagged error, suggested correction, and a direct link collapsed a multi-step manual lookup to about 5 min (conservative estimate).

Detection coverage moved from near 0% (customer-complaint only) to daily automated scanning of 25,000+ objects.

TERMS USED IN THIS CASE STUDY

DMAIC: Define, Measure, Analyze, Improve, Control - the Six Sigma improvement cycle

DPMO: defects per million opportunities. Each course had 6 checks, so each course counts as 6 opportunities

Sigma level (σ): a capability score calculated from DPMO; higher means fewer defects

LMS: learning management system

Lean Six Sigma: a method for removing waste (Lean) and reducing defects (Six Sigma)

SPC: statistical process control - tracking a process over time to separate normal variation from real problems

u-chart: a control chart for defect rates when the amount checked varies from reading to reading

CL: center line, the process average

UCL: upper control limit, 3 sigma above the center line; a point above it signals a special cause

Kaizen / Poka-Yoke: continuous improvement / mistake-proofing